



10.8.6 Wrap-Up: Cool Down

Anthony is conducting research to determine how students at his school get to school each day (biking, walking, car, bus, etc.). Because there are many students in his school, he will be conducting a survey of a random sample of students to make predictions about the overall population.

1. Select all the options that would produce a random sample to represent the overall population of the school.
 - ☐ Anthony will get to school before everyone else and survey the first 50 people who walk in the doors.
 - ☐ Anthony will survey everyone at his jazz band practice.
 - ☐ Anthony will survey two randomly chosen students in every homeroom.
 - ☐ Anthony will assign each student in the school a number, randomly choose 50 numbers, and survey those students.
 - ☐ Anthony will survey every third student he sees on his walk to school.



Unit 10, Lesson 9: Making Population Predictions – Part 1



Benchmark

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.



Learning Target

- ☐ I can use proportional relationships to make predictions about a population.



10.9.1 Warm-Up: Notice and Wonder

Burgers	Chips	Fries
Deluxe Burger	\$6.90	\$7.70
Deluxe Cheese	\$7.70	\$8.45
Bacon Burger	\$7.95	\$8.65
Bacon Cheese	\$8.25	\$8.95
Double Burger	\$8.35	\$9.30
Double Cheese	\$9.25	\$9.95
Double Bacon Cheese	\$9.50	\$10.25
Cheese Garden Burger	\$8.25	\$8.85
Mushroom Swiss	\$8.25	\$8.95
Jalapeno Pepper Jack	\$8.25	\$8.95
BBQ Western Bacon	\$9.20	\$9.85
Pastrami Burger	\$9.20	\$9.85
Power Pac	\$10.45	\$11.05
Cheese Power Pac	\$11.25	\$11.70

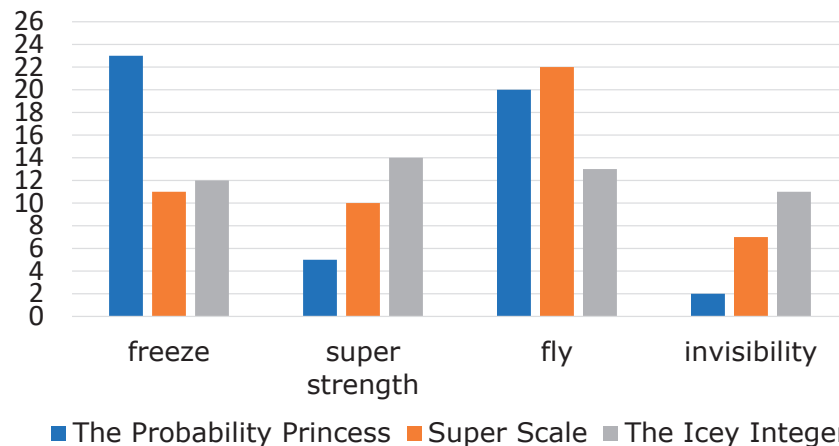
- What are two things you notice about this menu?
- What is one thing you wonder?
- What is a suggestion you could give to the owners about their pricing?



10.9.2 Guided Instruction: Predicting How Many

Results of a random sample can be used to make predictions. The results from a sample of 50 readers of each comic book are shown below.

**Superpowers Comic Readers
Would Like the New Hero to Have**



Suppose the author of *The Probability Princess*, Ashley Grant, is trying to make decisions about the future of the series. She really wants the new hero to have the ability to fly, but she knows there is a larger percentage of readers who want the new hero to have freeze power. As long as the new series can sell at least 3,600 issues in the next five years, Ashley will be able to stay in business.

- Ashley determines ____ readers of *The Probability Princess* responded to the survey. She then finds the percentage of the 50 readers surveyed who want the new hero to have the ability to fly.

$$\frac{\boxed{20}}{50} = \frac{40}{100} = \boxed{40} = 40\%$$

Over the past five years, *The Probability Princess* has had 10,000 loyal readers. Using three different strategies, Ashley uses the data to predict the number of those readers who want to see the new hero have the ability to fly. In each method, she predicts that 4,000 of the 10,000 readers want this.

Equivalent Fraction	Scale Factor	Percent
The fraction of readers who want the new hero to fly is $\frac{20}{50}$. Write equivalent fractions until the denominator shows the population total (in this case 10,000). $\frac{20}{50} = \frac{40}{100} =$ $\frac{400}{1,000} = \frac{4,000}{10,000}$	The fraction of readers who want the new hero to fly is $\frac{20}{50}$. Divide the population by the sample to find the scale factor needed. Then, multiply both the numerator and denominator by this scale factor to write an equivalent fraction. $10,000 \div 50 = 200$ $\frac{20}{50} \times \frac{200}{200} = \frac{4,000}{10,000}$	The fraction of readers who want the new hero to fly is $\frac{20}{50}$. Find the percent of the sample for the given condition. Then multiply the population by the percent. $20 \div 50 = 0.4 \text{ or } 40\%$ $10,000 \times 0.4 = 4,000$ OR $40\% = \frac{40}{100} = \frac{2}{5}$ $10,000 \times \frac{2}{5} = 4,000$

2. Explain whether one of Ashley's strategies seems to be better than the others for solving this particular problem.

3. What is the benefit of solving a problem using two different strategies?

4. Next, Ashley finds the percent of the 10,000 issues they need to sell in the next five years. Complete the work shown below.

$$\frac{3,600}{10,000} = \frac{\boxed{}}{100} = 0.36 = \boxed{}\%$$

Ashley estimates that if the new hero has the ability to fly, at least 4,000 readers will continue to read and purchase *The Probability Princess*. She estimates that this will lead to each of those 4,000 readers purchasing at least one comic. She also thinks it's likely that readers who did not want the new hero to have the ability to fly may still buy issues in the next five years. Because 4,000 is greater than the 3,600 issues they need to sell to stay in business, Ashley decides to give the next hero the ability to fly.

5. Explain why you agree or disagree with Ashley's decision.



10.9.3 Exploration: Reaction Times

The track coach at Yulee High School in Nassau County, FL needs a student whose reaction time is less than 0.4 seconds to help out with timers at track meets. All the twelfth graders in the school measured their reaction times.

A random sample of 32 students' reaction times has been recorded in the table.

Reaction Times of Less Than 0.4 Seconds	Reaction Times of 0.4 Seconds or More
22	10

1. What fraction of the students in the random sample have a reaction time less than 0.4 seconds?
2. Explain or show how to estimate the percentage of selected twelfth-graders from this school who had a reaction time of less than 0.4 seconds.
3. There are 125 twelfth graders at this school. How many of the twelfth graders are expected to have a reaction time of less than 0.4 seconds?
4. Suppose another group in your class comes up with a different estimate than yours for the previous question. What is another estimate that would be **reasonable**?
5. Explain what kind of an estimate would be considered **unreasonable**.



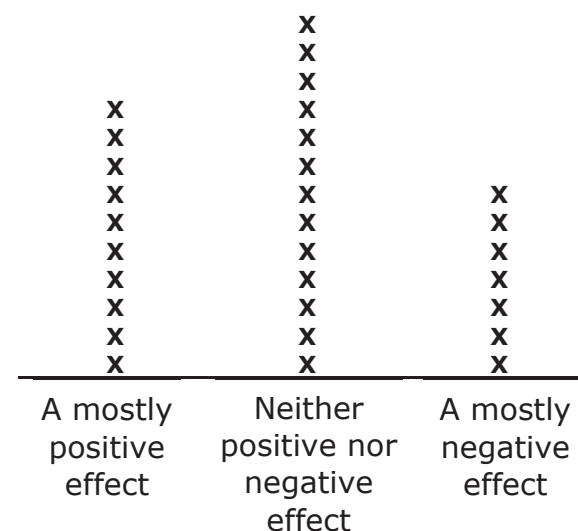
10.9.4 Guided Practice: Making Population Predictions

1. Taylor attends Murdock Middle School in Port Charlotte, Florida. She wonders how many students at her school would dye their hair blue if their parents would let them. She surveyed a random sample of 10 students at her school, and 2 of them said they would. Kiran didn't think Taylor's estimate was very accurate, so he surveyed a random sample of 100 students, and 17 of them said they would.
 - a. Complete the table below to compare the accuracy of Taylor and Kiran's random samples. Be sure to show your work or explain your thinking.

	Taylor	Kiran
Fraction of students surveyed who would dye their hair blue		
Percentage of students surveyed who would dye their hair blue		
Predicted number of the 500 students in the school who would dye their hair blue		

- b. Explain whose random sample likely produced the most accurate estimate.

2. Ms. Jeffrey teaches seventh grade in Miami, Florida. She noticed many of her students talked about social media and wondered about what her students considered to be the impact of all their online activity. She conducted a random survey of 30 seventh-grade students asking them how they describe the effect of social media use on people their own age. The results are shown in the line plot below.



According to a 2018 Pew Research Center study of American teenagers, 24% of teen respondents said they believed social media has a mostly negative effect on people their own age.

- a. Explain how this compares to the percentage of Ms. Jeffrey's seventh-grade students who described social media as having a mostly negative effect.

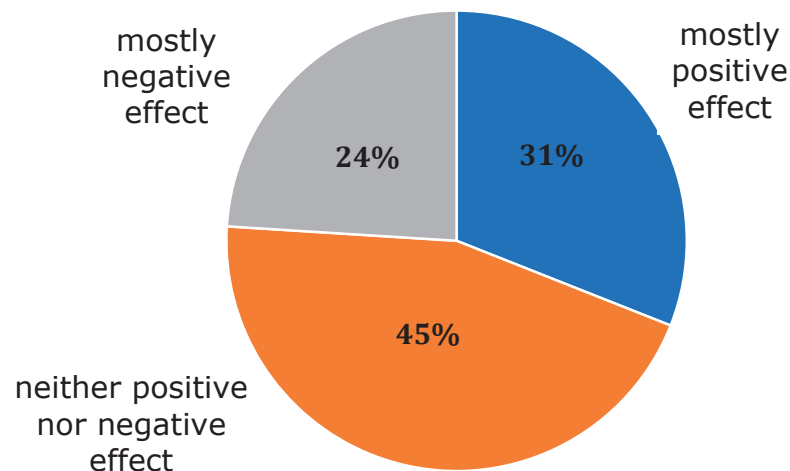
There were about 1,546,397 teenagers living in the state of Florida in 2018.

- b. Predict the number of teenagers in the state of Florida who considered the effect of social media as mostly positive based on the data Ms. Jeffrey collected.
- c. Predict the number of teenagers in the state of Florida who considered the effect of social media as neither positive nor negative based on the data Ms. Jeffrey collected.
- d. Explain at least two factors that could account for differences in these estimates from the actual number of students who would have described the effect of social media in these ways.

Why are they different?	
1.	
2.	

The results of the 2018 Pew Research Study are shown in the circle graph.

Teen Views on the Effect of Social Media on People Their Age



- e. The percentage of teens who describe social media as having a positive effect on people their age is ☐ less than ☐ equal to ☐ greater than the percent of students who responded the same in Ms. Jeffrey's survey.

Write an explanation of why this may have happened.



10.9.5 Wrap-Up: Growth Mindset

Complete the following sentences based on today's learning target:

I can use proportional relationships to make predictions about a population.

1. A barrier to my learning today was ...

2. I will try to overcome this by ...



Unit 10, Lesson 10: Making Population Predictions – Part 2



Benchmark

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.



Learning Target

- ☐ I can use proportional relationships to make predictions about a population.